전산화학 5 장 연습문제 풀이 (2010년 1 학기)

1. **Problem 1.**

(g)

$$\int \frac{RT}{p} dp = RT \ln p + C.$$

(i)

$$\int \frac{Q^2}{r^2} dr = -\frac{Q^2}{r} + C.$$

(j)

$$\int \cos(2\pi W t) dt = \frac{1}{2\pi W} \sin(2\pi W t) + C.$$

2. **Problem 2.**

(c)

(e)

$$\begin{aligned} \int \sin^2\left(\frac{N\pi x}{A}\right) dx &= \int \frac{1}{2}\left(1 - \cos\left(\frac{2N\pi x}{A}\right)\right) dx, \quad (\sin^2 x = \frac{1}{2}(1 - \cos 2x)) \\ &= \frac{1}{2}x - \frac{A}{4N\pi} \sin\left(\frac{2N\pi x}{A}\right) + C. \end{aligned}$$

(f)

$$\begin{aligned} \int \sin^2\left(\frac{N\pi x}{A}\right) x dx &= \int \frac{1}{2}\left(1 - \cos\left(\frac{2N\pi x}{A}\right)\right) x dx, \\ &= \int \left(\frac{1}{2}x - \frac{1}{2}x\cos\left(\frac{2N\pi x}{A}\right)\right) dx, \\ &= \frac{1}{4}x^2 - \frac{1}{2}\int x \cos\left(\frac{2N\pi x}{A}\right) dx. \qquad (1) \end{aligned}$$

If $u = x$ and $v' = \cos\left(\frac{2N\pi x}{A}\right) dx$, then $u' = dx$ and $u = \frac{A}{2N\pi} \sin\left(\frac{2N\pi x}{A}\right)$ Eq. (1) is given by

$$\begin{aligned} \frac{1}{4}x^2 - \int x \cos\left(\frac{2N\pi x}{A}\right) dx &= \frac{1}{4}x^2 - \frac{1}{2}\left\{ x\frac{A}{2N\pi} \sin\left(\frac{2N\pi x}{A}\right) - \int \frac{A}{2N\pi} \sin\left(\frac{2N\pi x}{A}\right)\right\}, \\ &= \frac{1}{4}x^2 - \frac{A}{4N\pi} x \sin\left(\frac{2N\pi x}{A}\right) + \frac{A}{4N\pi}\int \sin\left(\frac{2N\pi x}{A}\right), \\ &= \frac{1}{4}x^2 - \frac{A}{4N\pi} x \sin\left(\frac{2N\pi x}{A}\right) - \frac{A^2}{8N^2\pi^2} \cos\left(\frac{2N\pi x}{A}\right) + C. \end{aligned}$$

(g)

$$\int \left(\frac{\Delta H}{Rt^2}\right) dt = -\frac{\Delta H}{R}\left(-\frac{1}{t}\right) = \frac{\Delta H}{Rt}.$$

(o)

$$\int \left(\frac{\Delta H}{t^2} + \frac{A}{t} + \frac{B}{2} + \frac{C}{3}t\right) dt = -\frac{\Delta H}{t} + A \ln t + \frac{B}{2}t + \frac{C}{6}t^2 + D.$$

(r) If $u = r$ and $v' = e^{-ar}dr$, then $u' = dr$ and $v = -\frac{1}{a}e^{-ar}$.

$$\int re^{-ar}dr = -\frac{1}{a}re^{-ar} + \frac{1}{a}\int e^{-ar}dr = -\frac{1}{a}re^{-ar} - \frac{1}{a^2}e^{-ar}.$$

(t)

$$\int \frac{d(a)}{(a)} = -\int K dt \Rightarrow \ln a = -kt + C.$$

3. **Problem 3.**

(a)

$$\begin{aligned}\int_{T_1}^{T_2}\left(a + bT + cT^2 + \frac{d}{T}\right)dT &= \left[aT + \frac{b}{2}T^2 + \frac{c}{3}T^3 + d\ln T\right]_{T_1}^{T_2}, \\ &= a(T_2 - T_1) + \frac{b}{2}(T_2^2 - T_1^2) + \frac{c}{3}(T_2^3 - T_1^3) + d(\ln T_2 - \ln T_1).\end{aligned}$$

(d)

$$\int_{T_1}^{T_2}\frac{DeltaH}{RT^2}aT = \left[-\frac{\Delta H}{RT}\right]_{T_1}^{T_2} = -\frac{\Delta H}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right).$$

(e)

$$\begin{aligned}\int_{V_1}^{V_2}\left(\frac{nRT}{V - nb} - \frac{n^2a}{V^2}\right)dV &= -\left[nRT\ln(V - nb) + \frac{n^2a}{V}\right]_{V_1}^{V_2}, \\ &= -nRT\ln(V_2 - b) + nRT\ln(V_1 - nb) - \frac{n^2a}{V_2} + \frac{n^2a}{V_1}, \\ &= nRT\ln\left(\frac{V_1 - nb}{V_2 - nb}\right) - \frac{n^2a}{V_2} + \frac{n^2a}{V_1}.\end{aligned}$$

(g) If $\frac{n\pi x}{a} = t$, $dx = \frac{a}{n\pi}dt$ and

$$\begin{aligned}\int_0^a x^2\sin^2\left(\frac{n\pi x}{a}\right)dx &= \left(\frac{a}{n\pi}\right)^3\int_0^{n\pi} t^2\sin^2 t dt \\ &= \left(\frac{a}{n\pi}\right)^3\int_0^{n\pi} t^2\frac{1}{2}\Big(1 - \cos(2t)\Big)dt \\ &= \frac{1}{2}\left(\frac{a}{n\pi}\right)^3\int_0^{n\pi}\Big(t^2 - t^2\cos(2t)\Big)dt \\ &= \frac{1}{2}\left(\frac{a}{n\pi}\right)^3\left\{\left[\frac{1}{3}t^3\right]_0^{n\pi} - \underbrace{\int_0^{n\pi} t^2\cos(2t)dt}_{(*)}\right\} \qquad (2)\end{aligned}$$

Using intergration by parts, if $u = t^2$ and $v' = \cos(2t)dt$, then $u' = 2tdt$, $v =$

$\frac{1}{2}\sin(2t)$, and (*) becomes

$$\begin{aligned}\int_0^{n\pi} t^2\cos(2t)dt &= \left[\frac{1}{2}t^2\sin(2t)\right]_0^{n\pi} - \int_0^{n\pi} t\sin(2t)dt,\\ &= -\int_0^{n\pi} t\sin(2t)dt, \end{aligned} \tag{3}$$

Using intergration by parts again, if $u = t$ and $v' = \sin(2t)dt$, then $u' = dt$, $v = -\frac{1}{2}\cos(2t)$ and Eq. (2) becomes

$$\begin{aligned}\int_0^{n\pi} t^2\cos(2t)dt &= -\int_0^{n\pi} t\sin(2t)dt,\\ &= \left[\frac{1}{2}t\cos(2t)\right]_0^{n\pi} - \frac{1}{2}\int_0^{n\pi}\cos(2t)dt,\\ &= \frac{1}{2}n\pi - \frac{1}{2}\left[\frac{1}{2}\sin(2t)\right]_0^{n\pi},\\ &= \frac{1}{2}n\pi \end{aligned} \tag{4}$$

Inserting Eq. (4) into Eq. (2) leads to

$$\begin{aligned}\int_0^a x^2\sin^2\left(\frac{n\pi x}{a}\right)dx &= \frac{1}{2}\left(\frac{a}{n\pi}\right)^3\left(\frac{1}{3}(n\pi)^3 - \frac{1}{2}n\pi\right\},\\ &= \frac{1}{6}a^3 - \frac{1}{4}\frac{a^3}{n^2\pi^2}. \end{aligned}$$

(i) Using the integration by part, $u = r$ and $v' = e^{-2r/a_0}$, then $u' = dr$, $v = -\frac{a_0}{2}e^{-2r/a_0}$, and

$$\begin{aligned}\int_0^\infty e^{-2r/a_0}rdr &= \left[-\frac{a_0}{2}re^{-2r/a_0}\right]_0^\infty + \frac{a_0}{2}\int_0^\infty e^{-2r/a_0}dr,\\ &= -\left(\frac{a_0}{2}\right)^2\left[e^{-2r/a_0}\right]_0^\infty = \left(\frac{a_0}{2}\right)^2. \end{aligned}$$

(j) Using the integration by part, $u = v^2$ and $w' = ve^{-mv^2/2kT}dv$, then $u' = 2vdv$, $w = -\frac{kT}{m}e^{-mv^2/2kT}$.

$$\begin{aligned}\int_0^\infty e^{-mv^2/2kT}v^3dv &= \left[-\frac{kT}{m}v^2e^{-mv^2/kT}\right]_0^\infty + \frac{kT}{m}\int_0^\infty 2ve^{-mv/2kT}dv,\\ &= -2\left(\frac{kT}{m}\right)^2\left[e^{-mv/2kT}\right]_0^\infty = 2\left(\frac{kT}{m}\right)^2. \end{aligned}$$

(k) Let $t = J^2 + J$, then $dt = (2J+1)dJ$.

$$\int_0^\infty (2J+1)e^{-a(J^2+J)}dJ = \int_0^\infty e^{-at}dt = \frac{1}{a}$$

4. **Problem 5.**

(c) Let $u = \ln x$ and $v' = dx$, then $u' = \frac{1}{x}dx$ and $v = x$

$$\int \ln x dx = x \ln x - \int dx = x \ln x - x.$$

Thus,

$$\int\int y \ln x dx dy = \int y dy \int \ln x dx = \frac{1}{2}y^2(x \ln x - x) + C$$

(d) Let $u = x^2$ and $v' = e^{2x}dx$, then $u' = 2xdx$ and $v = \frac{1}{2}e^{2x}$

$$\begin{aligned}
\int x^2 e^{2x} dx &= \frac{1}{2}x^2 e^{2x} - \int x e^{2x} dx \\
&= \frac{1}{2}x^2 e^{2x} - \left\{ \frac{1}{2}x e^{2x} - \frac{1}{2}\int e^{2x} dx \right\} \\
&= \frac{1}{2}x^2 e^{2x} - \frac{1}{2}x e^{2x} + \frac{1}{4}e^{2x}.
\end{aligned}$$

Thus,

$$\begin{aligned}
\int\int\int x^2 \ln y e^{2x} dx dy dz &= \int x^2 e^{2x} dx \int \ln y dy \int dz, \\
&= \left(\frac{1}{2}x^2 e^{2x} - \frac{1}{2}x e^{2x} + \frac{1}{4}e^{2x} \right)(y \ln y - y)z + C.
\end{aligned}$$

(e)

$$\begin{aligned}
\int_0^{\pi/2} \int_0^2 r \cos\theta dr d\theta &= \int_0^2 r dr \int_0^{\pi/2} \cos\theta d\theta, \\
&= \left[\frac{1}{2}r^2 \right]_0^2 \left[\sin\theta \right]_0^{\pi/2}, \\
&= 2.
\end{aligned}$$

(f)

$$\begin{aligned}
\int_0^{2\pi} \int_0^{\pi} \int_0^V v^2 \sin\theta dv d\theta d\phi &= \int_0^V v^2 dv \int_0^{\pi} \sin\theta d\theta \int_0^{2\pi} d\phi, \\
&= \left[\frac{1}{3}v^3 \right]_0^V \left[-\cos\theta \right]_0^{\pi} \left[\phi \right]_0^{2\pi}, \\
&= \frac{4\pi}{3}V^3.
\end{aligned}$$

5. **Problem 8** Since $\Delta H = aT + \frac{b}{2}T^2 + \frac{c}{3}T^3 + d$ in Problem 7,

$$\left[\frac{\partial(\Delta G/T)}{\partial T} \right]_p = -\frac{\Delta H}{T^2} = -\frac{a}{T} - \frac{b}{2} - \frac{c}{3}T - \frac{d}{T^2}.$$

Thus,

$$\int \left[\frac{\partial(\Delta G/T)}{\partial T}\right]_p dT = \frac{\Delta G}{T} = \int \Big(-\frac{a}{T} - \frac{b}{2} - \frac{c}{3}T - \frac{d}{T^2}\Big) dT,$$
$$= -a\ln T - \frac{b}{2}T - \frac{c}{6}T^2 + \frac{d}{T} + e,$$

Finally,

$$\Delta G = -aT\ln T - \frac{b}{2}T^2 - \frac{c}{6}T^3 + d + eT.$$

6. **Problem 9** Let $t = \frac{\pi x}{L}$, then $dt = \frac{\pi}{L}dx$.

$$\begin{aligned}
\text{Probability} &= \frac{2}{L}\int_{L/2-0.05L}^{L/2+0.05L} \sin^2\frac{\pi x}{L}dx, \\
&= \frac{2}{L}\frac{L}{\pi}\int_{\pi/2-0.05\pi}^{\pi/2+0.05\pi} \sin^2 t dt, \\
&= \frac{2}{\pi}\int_{\pi/2-0.05\pi}^{\pi/2+0.05\pi} \frac{1}{2}\big(1-\cos(2t)\big)dt, \\
&= \frac{1}{\pi}\Big[t - \frac{1}{2}\sin(2t)\Big]_{\pi/2-0.05\pi}^{\pi/2+0.05\pi}, \\
&= \frac{1}{\pi}\Big(\frac{\pi}{2} + 0.05\pi - \frac{1}{2}\sin(\pi+0.1\pi) - \frac{\pi}{2} + 0.05\pi + \frac{1}{2}\sin(\pi-0.1\pi)\Big), \\
&= \frac{1}{\pi}(0.1\pi - 0.5\times(-0.309) + 0.5\times 0.309) \\
&= 0.1 + 0.309/\pi \\
&= 0.198.
\end{aligned}$$